Tutorial 5: Vector Autoregressive Model

Empirical Exercises

Auerbach and Gorodnichenko (2012) estimate the state-dependent effects of the government spending on the US economy using the smooth transition vector autoregressive (STVAR) model. In this exercise, we use their dataset to estimate the responses of output and taxes to the spending shock by means of linear VAR model. The dataset consists of three quarterly variables:

- G: the log of real government purchases.
- T: the log of real government receipts of direct and indirect taxes net of transfers to businesses and individuals.
- X: the log of real gross domestic product in chained 2000 dollars.

The sample covers period from 1948Q4 to 2008Q4.

1. Load the data of Auerbach and Gorodnichenko (2012). Produce a graph that shows government spending, taxes and GDP. Take care to annotate the graph properly.
2. Use the built-in function of Matlab to estimate a VAR(4) model of government spending, taxes and GDP (in this ordering).
3. Perform tests of Granger-noncausality for all three variables and report the results.
4. Assume $\mathbf{A}_0 = \mathbf{I}$ and $\mathbf{D}$ lower triangular. Compute the impulse responses to a one standard deviation spending shock and display them graphically. Add 90% bootstrap confidence intervals.
5. Rescale the impulse responses such that the spending shock instantaneously increases spending by 1 dollars. Add 90% bootstrap confidence intervals.
6. Change the specification to the ordering GDP, taxes, spending. Re-estimate and show the impulse responses to a one standard deviation spending shock. Compare with the previous model.